Phase 7 | Project Documentation

Sample Project Documentation

AI-Powered Travel Planner | Smart Situational Itinerary Engine

Live URL:	https://travel-planner-iota-plum.vercel.
GitHub:	github.com/itskian2007/travel-planner
Tech:	React + FastAPI + PostgreSQL + Vercel
Team:	itskian2007

1. Project Abstract

The AI-Powered Travel Planner is a full-stack web application that automates the creation of personalised 10-day travel itineraries for 20 global destinations. The system incorporates five preference dimensions (mood, budget, dietary, traveller persona, accessibility) to generate unique itineraries. Expert-curated Day 1-3 content is combined with a procedural engine for Days 4-10. All 304 landmark images are sourced from Wikipedia's verified image database with zero duplicates. Live at: https://travel-planner-iota-plum.vercel.app/

2. Project Specifications

Attribute	Value
Project Name	AI-Powered Travel Planner - Smart Situational Itinerary Engine
Domain	Travel Technology / AI-ML / Web Application
Frontend	React.js 18 + TailwindCSS + React Router v6
Backend	FastAPI (Python 3.11) + Uvicorn ASGI
Database	PostgreSQL 15 with SQLAlchemy ORM
Authentication	JWT (JSON Web Tokens) with bcrypt password hashing
Deployment	Vercel (Frontend) + Cloud (Backend) + Managed PostgreSQL
External API	Wikipedia REST API (landmark image sourcing)
Destinations	20 global cities across 6 continents
Total Images	304 unique, verified, non-duplicate landmark photographs
Itinerary Days	10 days per destination (Day 1-3 curated + Day 4-10 procedural)
Live URL	https://travel-planner-iota-plum.vercel.app/
GitHub	https://github.com/itskian2007/travel-planner

3. User Manual - Step-by-Step Guide

Step 1: Visit the Website

Open any browser and navigate to https://travel-planner-iota-plum.vercel.app/

The landing page presents the platform overview and a "Get Started" call-to-action.

Step 2: Register / Login

Click "Register" to create a new account (name, email, password).

Existing users click "Login" and enter credentials. A JWT token is issued on success.

Step 3: Set Your Preferences

On the Home page, use the Filter Panel to select:

- Mood: Adventure / Cultural / Relaxation / Romance
- Budget: Budget / Mid-Range / Luxury
- Dietary: Vegetarian / Vegan / Halal / Non-Veg / Gluten-Free
- Traveller Persona: Solo / Family / Couple / Group
- Accessibility needs (if any)

Step 4: Browse Destinations

The destination grid updates in real-time based on your filters. Browse 20 destinations with verified landmark images. Click any destination card to explore it further.

Step 5: View Your Itinerary

Click "Generate Itinerary" to create your personalised 10-day plan.

Days 1-3 feature expert-curated activities; Days 4-10 are procedurally generated.

Each day shows 3-4 attractions with Wikipedia-verified photographs.

Step 6: Explore Day-by-Day

Expand each day card to see: attractions, dining recommendations, estimated costs, local tips, and high-resolution landmark images for every activity.